elastic

Python powered Cibersecurity Lab

CALDERA

Automated adversary emulation

November 2026

PyconES2026

15:30

Program

Setup

The castle with all the required tools and infrastructure

Elasticsearch

The SIEM: logs, timelines and reports.

MITRE Caldera

Automated adversarial emulation

Correlación de eventos

Tracing behaviour and events

DISCLAIMER

Esta presentación tiene fines exclusivamente educativos y de investigación en seguridad. Todas las demostraciones, técnicas y escenarios mostrados se realizan en entornos controlados y autorizados. El contenido está orientado a la evaluación de riesgos, validación de controles de seguridad y mejora de las capacidades defensivas. No se promueve ni autoriza el uso de estas técnicas contra sistemas, redes o datos sin consentimiento explícito de sus propietarios.

SETUP



UBUNTU + ELASTIC

The center of the castle



UBUNTU VICTIM

The playground



EVIL MACHINE

Attacker

Setup







Docker

Caldera

Elastic

Elasticsearch

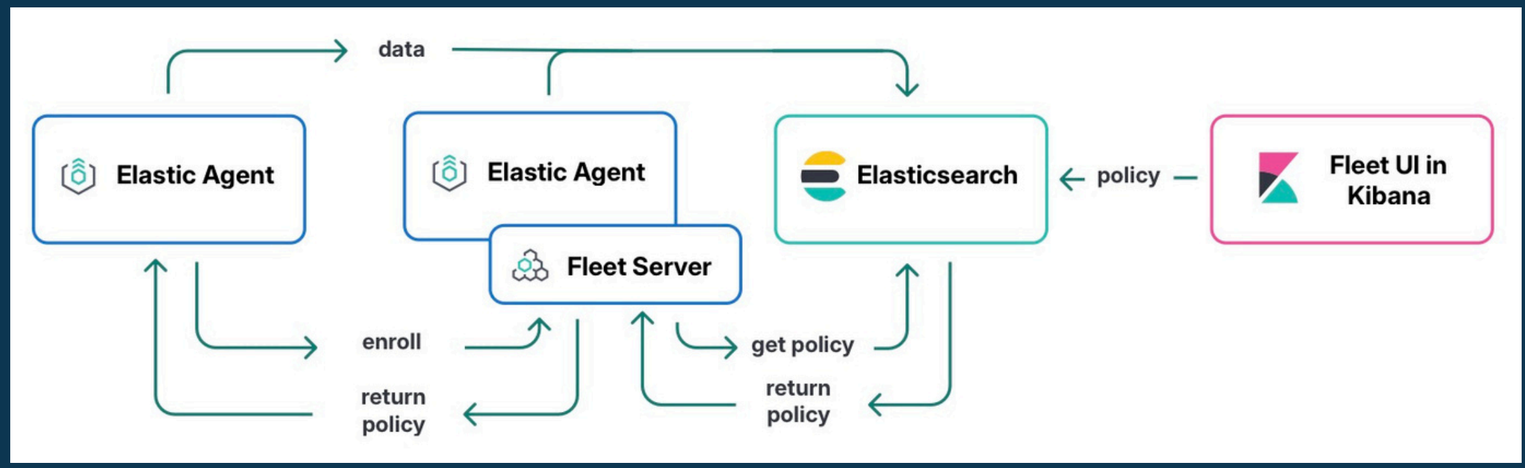
Security information Event Management

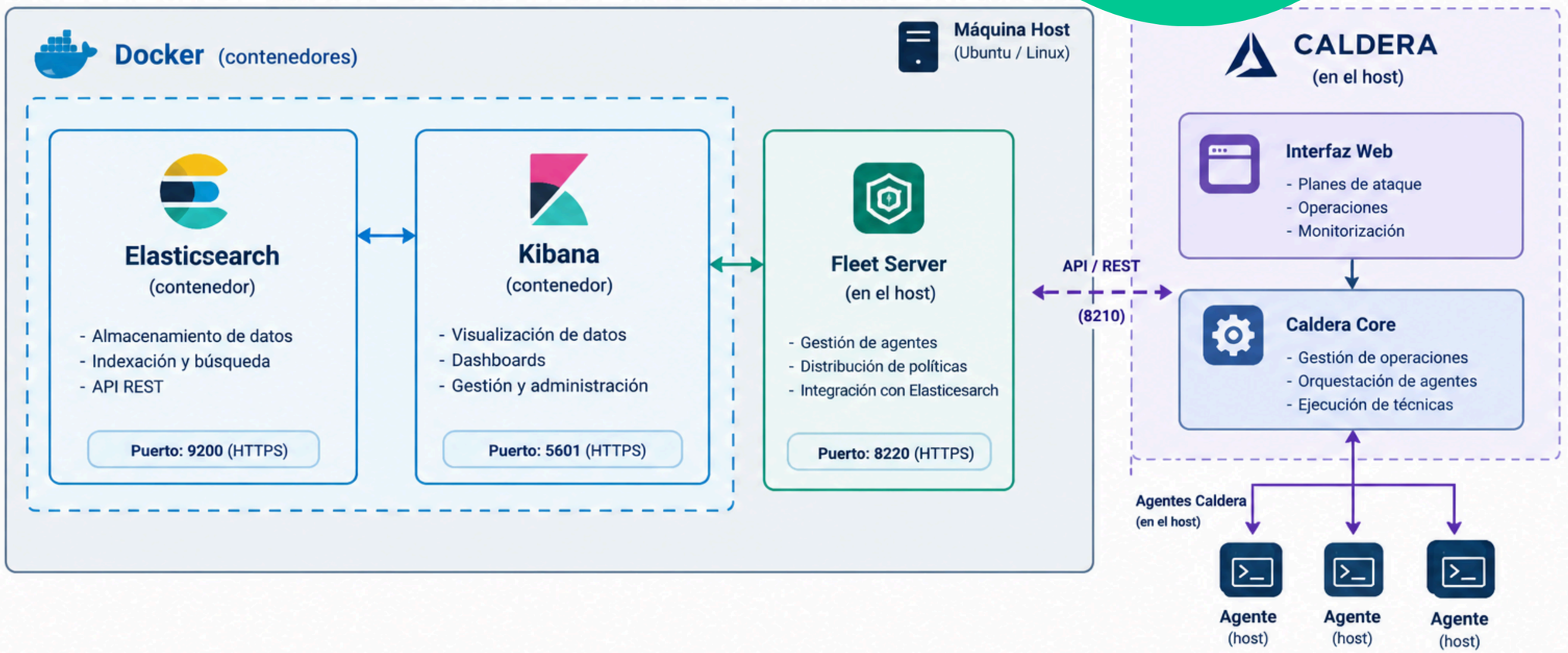
Stack

- Elastic server
- Kibana
- Fleet server

Integrations:

- Defend
- Audit Manager
- System audit





Leyenda

- Comunicación entre servicios (dentro de Docker)
- Comunicación Caldera ↔ Fleet Server
- Comunicación Caldera ↔ Agentes
- Acceso de usuario (navegador)

Servicios en Docker

- Elasticsearch (9200)
- Kibana (5601)

En el host

- Fleet Server (8220)
- Caldera (Web + Core)
- Agentes de Caldera

Acceso

Navegador web

- **Kibana:** https://localhost:5601
- **Caldera:** https://localhost:8888 (puerto según configuración)

Arquitectura del entorno de laboratorio

CALDERA™ provides an intelligent, automated offensive security testing (red teaming) platform that reduces the workload associated with routine security assessments, enabling security teams to focus on more critical and complex challenges.

- Minimum instalation time
- Go, Node, env
- Command and Control
- Sandcat Remote Access Trojan (RAT)





...a globally accessible, knowledge base of cyberattacker behaviors. Built by the MITRE Corporation, it acts as a standardized dictionary and playbook used by cybersecurity professionals to map, detect, and mitigate threats


naissance
echniques
Resource Development
9 techniques
Acquire Access
Acquire Infrastructure (8)
Compromise Accounts (3)
Compromise Infrastructure (8)
Develop Capabilities (4)
Establish Accounts (3)
Generate Content (2)
Obtain Capabilities (7)
Stage Capabilities (6)
Initial Access
11 techniques
Content Injection
Drive-by Compromise
Exploit Public-Facing Application
External Remote Services
Hardware Additions
Phishing (4)
Replication Through Removable Media
Supply Chain Compromise (3)
Trusted Relationship
Valid Accounts (4)
Wi-Fi Networks
Execution
20 techniques
BITS Jobs
Cloud Administration Command
Command and Scripting Interpreter (13)
Container Administration Command
Deploy Container
ESXi Administration Command
Exploitation for Client Execution
Hijack Execution Flow (12)
Input Injection
Inter-Process Communication (3)
Native API
Poisoned Pipeline Execution
Scheduled Task/Job (5)
Serverless Execution
Shared Modules
Software Deployment Tools
System Services (3)
Trusted Developer Utilities Proxy Execution (3)
User Execution (5)
Windows Management Instrumentation
Persistence
22 techniques
Account Manipulation (7)
BITS Jobs
Boot or Logon Autostart Execution (14)
Boot or Logon Initialization Scripts (5)
Cloud Application Integration
Compromise Host Software Binary
Create Account (3)
Create or Modify System Process (5)
Event Triggered Execution (18)
Exclusive Control
External Remote Services
Implant Internal Image
Modify Authentication Process (9)
Modify Registry
Office Application Startup (6)
Power Settings
Pre-OS Boot (5)
Scheduled Task/Job (5)
Server Software Component (6)
Software Extensions (2)
Privilege Escalation
13 techniques
Abuse Elevation Control Mechanism (6)
Access Token Manipulation (5)
Account Manipulation (7)
Boot or Logon Autostart Execution (14)
Boot or Logon Initialization Scripts (5)
Create or Modify System Process (5)
Domain or Tenant Policy Modification (2)
Escape to Host
Event Triggered Execution (18)
Exploitation for Privilege Escalation
Process Injection (12)
Scheduled Task/Job (5)
Valid Accounts (4)
Stealth
30 techniques
Access Token Manipulation (5)
BITS Jobs
Build Image on Host
Debugger Evasion
Delay Execution
Deobfuscate/Decode Files or Information
Direct Volume Access
Execution Guardrails (2)
Exploitation for Stealth
Hide Artifacts (14)
Hijack Execution Flow (12)
Indicator Removal (8)
Indirect Command Execution
Masquerading (12)
Obfuscated Files or Information (18)
Pre-OS Boot (5)
Process Injection (12)
Reflective Code Loading
Rootkit
Selective Exclusion
Social Engineering (2)
System Binary Proxy Execution (14)
Defense Impairment
18 techniques
Disable or Modify System Firewall (3)
Disable or Modify Tools (6)
Domain or Tenant Policy Modification (2)
Downgrade Attack
Exploitation for Defense Impairment
File and Directory Permissions Modification (2)
Modify Authentication Process (9)
Modify Cloud Compute Infrastructure (5)
Modify Cloud Resource Hierarchy
Modify Registry
Modify System Image (2)
Network Boundary Bridging (1)
Plist File Modification
Prevent Command History Logging
Rogue Domain Controller
Safe Mode Boot
Subvert Trust Controls (6)
Weaken Encryption (2)
Credential Access
17 techniques
Adversary-in-the-Middle (4)
Brute Force (4)
Credentials from Password Stores (6)
Exploitation for Credential Access
Forced Authentication
Forge Web Credentials (2)
Input Capture (4)
Modify Authentication Process (9)
Multi-Factor Authentication Interception
Multi-Factor Authentication Request Generation
Network Sniffing
OS Credential Dumping (8)
Steal Application Access Token
Steal or Forge Authentication Certificates
Steal or Forge Kerberos Tickets (5)
Steal Web Session Cookie
Unsecured Credentials (8)
Discovery
34 techniques
Account Discovery (4)
Application Window Discovery
Browser Information Discovery
Cloud Infrastructure Discovery
Cloud Service Dashboard
Cloud Service Discovery
Cloud Storage Object Discovery
Container and Resource Discovery
Debugger Evasion
Device Driver Discovery
Domain Trust Discovery
File and Directory Discovery
Group Policy Discovery
Local Storage Discovery
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Lateral Movement
9 techniques
Exploitation of Remote Services
Internal Spearphishing
Lateral Tool Transfer
Remote Service Session Hijacking (2)
Remote Services (8)
Replication Through Removable Media
Software Deployment Tools
Taint Shared Content
Use Alternate Authentication Material (4)
Collection
17 techniques
Adversary-in-the-Middle (4)
Archive Collected Data (3)
Audio Capture
Automated Collection
Browser Session Hijacking
Clipboard Data
Data from Cloud Storage
Data from Configuration Repository (2)
Data from Information Repositories (6)
Data from Local System
Data from Network Shared Drive
Data from Removable Media
Data Staged (2)
Email Collection (3)
Input Capture (4)
Screen Capture
Video Capture
Command and Control
18 techniques
Application Layer Protocol (5)
Communication Through Removable Media
Content Injection
Data Encoding (2)
Data Obfuscation (3)
Dynamic Resolution (3)
Encrypted Channel (2)
Fallback Channels
Hide Infrastructure
Ingress Tool Transfer
Multi-Stage Channels
Non-Application Layer Protocol
Non-Standard Port
Protocol Tunneling
Proxy (4)
Remote Access Tools (3)
Traffic Signaling (2)
Web Service (3)
Exfiltration
9 techniques
Automated Exfiltration (1)
Data Transfer Size Limits
Exfiltration Over Alternative Protocol (3)
Exfiltration Over C2 Channel
Exfiltration Over Other Network Medium (1)
Exfiltration Over Physical Medium (1)
Exfiltration Over Web Service (4)
Scheduled Transfer
Transfer Data to Cloud Account
Impa
15 techn
Account A Removal
Data Destruct
Data Encry Impact
Data Manipul
Defacem
Disk Wip
Email Bom
Endpoin of Servic
Financial T
Firmware Corruption
Inhibit Sys Recovery
Network of Servic
Resourc Hijackin
Service St
System Shutdown/

Require ments

Resource	Elstc
RAM	**16 GB**
Disco	**500 GB**
Caldera	**5.3.0**
Elastic	**9.5.3**
Ubuntu	**26.04**
Python	**3.11.15**

PRACTICE

- Virtualization
- Terminal
- Linux (Ubuntu)
- Compatibility packet installation
- Networking
- http y transport and CA certificates
- Logs reading
- Cibersecurity
- Directories
- OS instalation

LET’S START!!

Let’s find how the test generate traces in our SIEM.



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